concentrated into a minuscule region of space called a *singularity*.[16] That singularity, roughly 10^{-33} centimeter in size (a hundred billion billion times smaller than an atomic nucleus), should be surrounded by pure emptiness, aside from the tenuous interstellar gas that is falling inward now and the radiation the gas emits. There should be near emptiness from the singularity out to the horizon, and near emptiness from the horizon out to your starship.

Box P.1

Power Notation for Large and Small Numbers

In this book I occasionally will use "power notation" to describe very large or very small numbers. Examples are 5×10^{6}, which means five million, or 5,000,000, and 5×10^{-6}, which means five millionths, or 0.000005.

In general, the power to which 10 is raised is the number of digits through which one must move the decimal point in order to put the number into standard decimal notation. Thus 5×10^{6} means take 5 (5.00000000) and move its decimal point rightward through six digits. The result is 5000000.00. Similarly, 5×10^{-6} means take 5 and move its decimal point leftward through six digits. The result is 0.000005.

The singularity and the stellar matter locked up in it are hidden by the hole's horizon. However long you may wait, the locked-up matter can never reemerge. The hole's gravity prevents it. Nor can the locked-up matter ever send you information, not by radio waves, or light, or X-rays. For all practical purposes, it is completely gone from our Universe. The only thing left behind is its intense gravitational pull, a pull that is the same on your 1-million-kilometer orbit today as before the star imploded to form the hole, but a pull so strong at and inside the horizon that nothing there can resist it.

"What is the distance from the horizon to the singularity?" you ask yourself. (You choose not to measure it, of course. Such a measurement would be suicidal; you could never escape back out of the horizon to report